MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, SCHOTTKY RECTIFIERS,
JANTX AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for schottky rectifiers intended for use in power surface mount applications. Two levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1 (surface mount).

1.3 Maximum ratings.

Column 1	Column 2	Column 3	Column 4	Column 5		Column 6	
Types 1/	Peak inverse voltage	I_O 2/ T_C = +100°C	I_{FSM} 3/	Maximum 4/ thermal resistance		T_{STG}	T_{OP}
	V dc	A dc	A(pk)	°C/W 5/	°C/W 6/	°C	°C
M19500/619-13Z	30	110	1,000	0.20	0.45	-65 to +175	-65 to +150
M19500/619-23Z	45	150	1,000	0.20	0.45		-65 to +175
M19500/619-33Z	100	120	900	0.20	0.45		-65 to +175
M19500/619-12Z	30	60	900	0.35	0.80		-65 to +150
M19500/619-22Z	45	75	900	0.35	0.80		-65 to +175
M19500/619-32Z	100	60	900	0.35	0.80		-65 to +175
M19500/619-21Z	45	10	300	0.85	1.40		-65 to +175
M19500/619-31Z	100	10	300	0.85	1.40		-65 to +175

1/ See 1.5 for type designation/lead configuration.
2/ Derate the I_O linearly to zero at 100°C to the maximum operating temperature.
3/ I_{FSM}, t_p = 8.3 ms, T_C = +25°C, V_R = PIV.
4/ Maximum thermal resistance junction to cathode surface.
5/ Thermal resistance value for package outline (YZ) (i.e., 11, 21, 31, 12, 22, 32, 13, 23, and 33).
6/ Thermal resistance value for package outline (YZ) (i.e., 14, 24, 34).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ELDT, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Primary electrical characteristics at T_C = 25°C.

Column 7	Column 8	Column 9	Column 10	Column 11	Column 12	Column 13	Column 14
Types	Maximum forward voltage per leg T_C = +25°C		Maximum forward voltage per leg T_C = +125°C		Maximum reverse current per leg at PIV mA		Maximum junction capacitance per leg 1 KHz ≤ f ≤ 1 MHz V_R = 5 V
	Volts	Amperes	Volts	Amperes	T_J = +25°C	T_J = +125°C	pF
M19500/619-13Z	0.53	110	0.43	110	7.5	400 1/	7,500
M19500/619-23Z	0.75	150	0.65	150	5.0	150	5,000
M19500/619-33Z	0.90	120	0.73	120	5.0	100	2,700
M19500/619-12Z	0.55	60	0.49	60	5.0	200 1/	3,700
M19500/619-22Z	0.74	75	0.70	75	5.0	45	2,600
M19500/619-32Z	0.90	60	0.83	60	1.5	20	1,400
M19500/619-21Z	0.55	10	0.49	10	2.0	15	1,000
M19500/619-31Z	0.75	10	0.63	10	1.0	10	600

1/ T_J = +110°C.

1.5 Type designation. The type designation used under this part shall be formulated as follows:

M19500/619-X Y Z
- Z — Package outline.
- Y — Package size.
- X — Voltage type.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S 19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figure 1 herein.

3.3.1 Base, lid, and lead finish. Base and lid material shall be Molybdenum. Lead finish shall be tin-plated or solder-dipped. When a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.3.2 Internal construction. Multiple chip construction shall not be permitted.

3.4 Marking. Devices shall be marked in accordance with MIL-S-19500, abbreviation of the part can be as follows JVM619-XYZ.

3.5 Electrostatic discharge protection. The devices covered by this specification shall be classified for ESD sensitivity in accordance with MIL-S-19500 and MIL-STD-750, method 1020, and shall be handled, marked, and packaged accordingly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.3 Screening (JANTX, AND JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurements
	JANTX and JANTXV levels
Thermal response 1/	MIL-STD-750, method 3101, (see 4.5.3)
9	Not applicable
10	MIL-STD-750, method 1038, test condition B, T_C = +100°C
11	Subgroup 2 of table I herein.
12	Not applicable
13	Subgroup 2 of table I herein; $\Delta I_R \leq 5.0$ mA or 100 percent of initial whichever is greater. $\Delta V_F \leq .05$ volts or ±100 percent of initial value, whichever is greater.

1/ Shall be performed anytime before screen 10.

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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500 and as specified herein. Group A inspection shall be performed on each sublot.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with 4.3, screen 13.

4.4.2.1 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

a. Subgroup 3: Condition for intermittent operation life are as follows: V_{RWM} = rated PIV, I_O set to maintain ΔT_C = +100°C, T_A = +25°C ±3°C; $t_{on} = t_{off}$ = 1 minute minimum for 2,000 cycles. No heat sink or forced air shall be permitted.

b. Subgroup 4: Condition for decap-internal visual (design verification), see 3.3.2.

c. Subgroup 5: Condition for thermal resistance, see 4.4.2.

NOTE: Subgroups 2, 3, and 6, electrical measurements, see 4.3, screen 13.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with 4.3, screen 13).

4.4.3.1 Group C inspection, table V of MIL-S-19500.

a. Subgroup 3: Condition for shock is 1,500 G's, 0.5 ms.

b. Subgroup 6: Condition for intermittent operation life are as follows: V_{RWM} = rated PIV, I_O set to maintain ΔT_C = +100°C, T_A = +25°C ±3°C; $t_{on} = t_{off}$ = 1 minute minimum for 6,000 cycles. No heat sink or forced air shall be permitted. (Test condition D).

NOTE: Subgroups 2, 3, and 6, electrical measurements, see 4.3, screen 13.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions for subgroup testing in table II herein. Electrical measurements shall be in accordance with 4.3, screen 13.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurement shall be conducted in accordance with test method 4081 of MIL-STD-750. The maximum limit of $R_{\theta JC}$(max) shall be in accordance with 1.3.

4.5.3 Thermal response. The ΔV measurement shall be performed in accordance with method 3101 of MIL-STD-750. The ΔV conditions (I_H and V_H) and maximum limit shall be derived by each vendor from the thermal response curves. The following parameter measurements shall apply:

a. Measuring current (I_M) . . . . . +100 mA.

b. Heating current (I_H) . . . . . 20 A minimum.

c. Heating time (t_H) . . . . . 100 ms.

d. Measurement time delay (t_{MD}) . . . . . +10 to 100 µs.

e. Sample window time (t_{SW}) . . . . . 5 µs maximum.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits 2/		Unit
	Method	Conditions 2/		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Forward voltage	4011	t_p = 300 µs I_F^p = Column 9	V_{FM}		Col 8	V (pk)
Reverse current	4016	V_R = Column 2	I_{R1}		Col 12	mA dc
Subgroup 3						
High temperature operation:		T_C = +125°C				
Reverse current	4016	V_R = Column 2	I_{R2}		Col 13	mA dc
Subgroup 4						
Capacitance (per leg)	4001	V_R = 5 V; 1 kHz ≤ f ≤ 1 MHz	C_J		Col 14	pF
Subgroup 5						
Surge current	4066	I_{FSM} = rated (Column 4), 10 surges of 8.3 ms each at 1 minute intervals; V_{RWM} = rated PIV (Column 2)				
Electrical measurements		See 4.3, screen 13				

1/ For sampling plan, see MIL-S-19500.
2/ See 1.3 and 1.4 for conditions and limits.

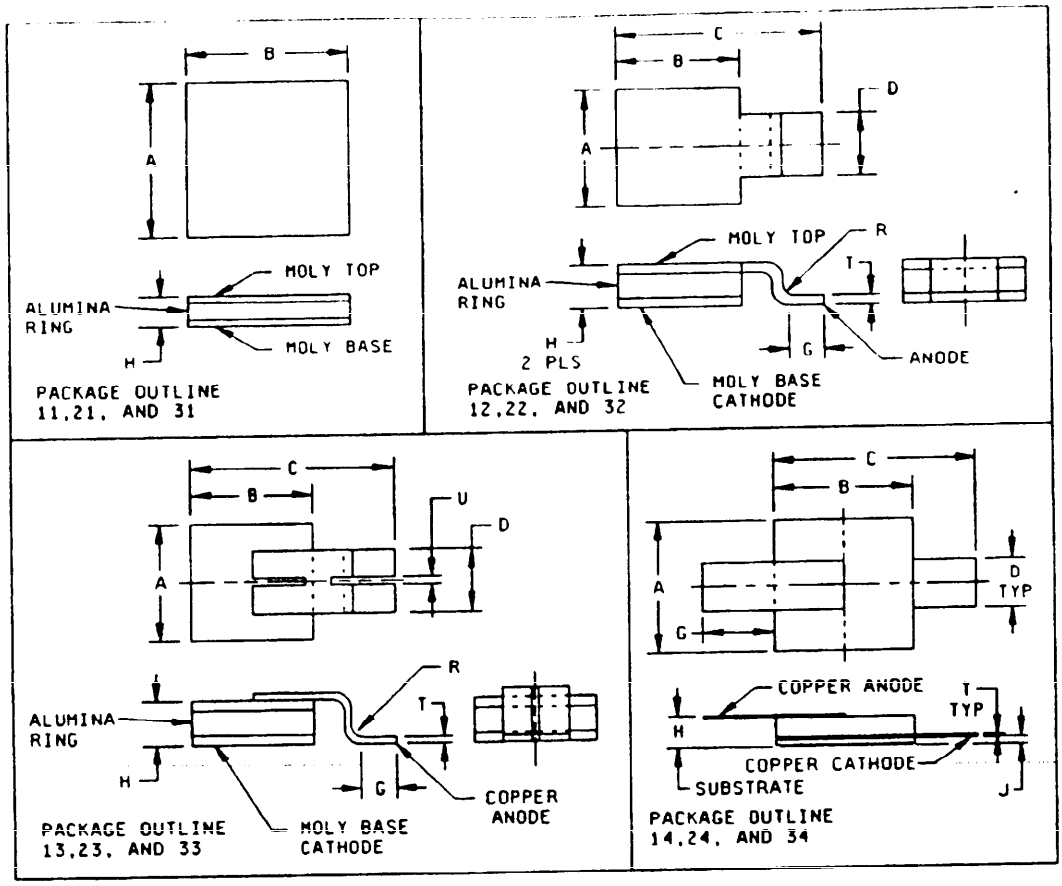
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TABLE II. Group E inspection (all quality levels) for qualification only. 1/

Inspection	MIL-STD-750		Sample size
	Method	Conditions	
Subgroup 1 Thermal shock Electrical measurement	1051	-55°C to +150°C, 500 cycles See 4.3, screen 13	22 c = 0
Subgroup 2 2/ Steady-state reverse bias Electrical measurement	1038	Test condition A; 1,000 hours; T_J = +150°C; V_R = 80-85 percent rated PIV. See 4.3, screen 13	22 c = 0
Subgroup 3 Not applicable			
Subgroup 4 2/ Thermal resistance, junction to lead	3101 or 4081	$R_{\Theta JC}$ = Column 5(°C/W maximum) See 4.5.2	11 c = 0
Subgroup 5 Not applicable			
Subgroup 6 ESD	1020		11 c = 0

1/ For initial design verification only (one time testing).
2/ A separate sample may be pulled for each test.

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Package outline (YZ)	A	B	C	D	G	H	J	R	T	U
11	0.225	0.225	-	-	-	0.075 max	-	-	-	-
21	0.315	0.315	-	-	-	0.075 max	-	-	-	-
31	0.410	0.410	-	-	-	0.075 max	-	-	-	-
12	0.225	0.225	0.375	0.125	0.060	0.075 max	-	0.020	0.020	-
22	0.315	0.315	0.465	0.170	0.060	0.075 max	-	0.020	0.020	-
32	0.410	0.410	0.550	0.200	0.060	0.075 max	-	0.020	0.020	-
13	0.225	0.225	0.375	0.150	0.060	0.090 max	-	0.020	0.015	0.015
23	0.315	0.315	0.465	0.150	0.060	0.090 max	-	0.020	0.015	0.015
33	0.410	0.410	0.560	0.150	0.060	0.090 max	-	0.020	0.015	0.015
14	0.225	0.225	0.405	0.150	0.190	0.138 max	0.037	-	0.012	-
24	0.315	0.315	0.495	0.150	0.190	0.138 max	0.037	-	0.012	-
34	0.410	0.410	0.590	0.150	0.190	0.138 max	0.037	-	0.012	-
Tolerances	±0.010	±0.010	±0.020	±0.010	±0.010	-	±0.005	±0.005	±0.002	±0.005

NOTE: All dimensions are in inches.

FIGURE 1. Physical dimensions.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Product assurance level and military Part or Identifying Number (PIN).

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, AV, MI, SM
Navy - AS, CG, MC
Air Force - 19, 85, 99

Preparing activity:
DLA - ES

(Project 5961-1454)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/619	2. DOCUMENT DATE (YYMMDD) 94/05/11

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, SCHOTTKY RECTIFIERS, JANTX AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

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c. ADDRESS (Include Zip Code) Defense Electronics Supply Center Attn: DESC-ELDT Dayton, Ohio 45444 5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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